HOMEWORK 4

1.

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 9 \\ -6 & -4 \end{bmatrix} \qquad c = \langle 2, 3, -6 \rangle \qquad d = \langle -1, 9, -4 \rangle$$

$$\vec{U}_1 = c = \langle 2, 3, -6 \rangle \qquad \text{proj}_c = \frac{d \cdot c}{c \cdot c} \vec{c} = \frac{-2 + 27 + 24}{4 + 9 + 36} \langle 2, 3, -6 \rangle$$

$$\vec{U}_2 = d - \text{proj}_d c \qquad = \frac{49}{49} \langle 2, 3, -6 \rangle = \begin{bmatrix} 2 \\ 3 \\ -6 \end{bmatrix}$$

$$\vec{U}_2 = \begin{bmatrix} -1 \\ 9 \\ -4 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \\ -6 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \\ 2 \end{bmatrix}$$

$$q_1 = \frac{U_1}{\|U_1\|} = \begin{bmatrix} 2/7 \\ 3/7 \\ -6/7 \end{bmatrix} \qquad q_2 = \begin{bmatrix} -3/7 \\ 6/7 \\ 2/7 \end{bmatrix}$$

$$Q = [q_1 \quad q_2] = \begin{bmatrix} 2/7 & -3/7 \\ 3/7 & 6/7 \\ -6/7 & 2/7 \end{bmatrix} \qquad \begin{aligned} A &= QR \\ R &= Q^T A \end{aligned}$$

$$\begin{bmatrix} 2/7 & 3/7 & -6/7 \\ -3/7 & 6/7 & 2/7 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 9 \\ -6 & -4 \end{bmatrix} = \begin{bmatrix} 7 & 7 \\ 0 & 7 \end{bmatrix}$$

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[signature]

2.

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 9 \\ -6 & -4 \end{bmatrix} = \begin{bmatrix} 2/7 & -3/7 \\ 3/7 & 6/7 \\ -6/7 & 2/7 \end{bmatrix} \begin{bmatrix} 7 & 7 \\ 0 & 7 \end{bmatrix} = QR$$

$$R\vec{x} = Q^T \vec{b}$$

$$\begin{bmatrix} 7 & 7 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2/7 & 3/7 & -6/7 \\ -3/7 & 6/7 & 2/7 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

by back substitution,

$$x_2 = -1/7$$

$$7x_1 + 7\left(-\frac{1}{7}\right) = -1 \rightarrow x_1 = 0$$

$$\vec{x} = \begin{bmatrix} 0 \\ 1/7 \end{bmatrix}$$

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3.a.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 6 \\ 0 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix} = P \cdot U$$

$\det(A) = \det(P) \cdot \det(U)$

$\det(P) = -1$

$\det(U) = 1 \cdot 4 \cdot 6 = 24$ (triangular matrix)

$\det(A) = -1 \cdot 24 = -24$

3.b.

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 5 & 7 \\ 1 & 2 & 3 & 100 \\ 3 & 5 & 8 & 11 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 5 & 7 \\ 1 & 2 & 3 & 100 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\det(B) = 0$

3.c.

$$C = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Diagonal matrice

$\det(C) = 2 \cdot 3 \cdot 6 \cdot 1 = 36$

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3.d.

$$D = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} = (-1)^{1+1} \cdot 1 \cdot 1 + (-1)^{1+2} \cdot 2 \cdot 2 + (-1)^{1+3} \cdot 3(-3)$$

$$= 1 - 4 + (-9) = -12$$

4.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 6 & 10 \end{bmatrix} \qquad b = \begin{bmatrix} 1 \\ 4 \\ 6 \end{bmatrix}$$

$$A_1 = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 8 \\ 6 & 6 & 10 \end{bmatrix} \Rightarrow \det(A_1) = 1 \cdot 2 - 2(-8) + 3(-6) = 0$$

$$A_2 = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 4 & 8 \\ 3 & 6 & 10 \end{bmatrix} \longrightarrow \det(A_2) = -4$$

$$A_3 = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 4 \\ 3 & 6 & 6 \end{bmatrix} \longrightarrow \det(A_3) = 3$$

$$\det(A) = 1 \cdot 2 - 2(-4) + 3(-3) = 1$$

$$x_i = \frac{\det(A_i)}{\det(A)} \qquad x_1 = 0, \ x_2 = -4, \ x_3 = 3$$

$$\vec{x} = \begin{bmatrix} 0 \\ -4 \\ 3 \end{bmatrix}$$

Sorry EK SUM

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